

On a novel supersymmetric connection between harmonic and isotonic oscillators

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Abstract

Beginning from the harmonic oscillator we obtain a one-parameter family of potentials whose spectra coincide with that of the isotonic oscillator. Diverse arguments concerning supersymmetric transformations, factorization procedures and Riccati equations are used in a systematic way. In particular we take advantage of the non-standard factorization of the Schrödinger equation which appears when considering the energy of the first excited level as reference. To be more specific, supersymmetry is neither exact nor spontaneously broken but realizes itself acting on wave functions vanishing at $x = 0$.

Supersymmetry represents in modern physics a new kind of symmetry, which, mixing both bosonic and fermionic degrees of freedom, provides a natural underlying structure in a wide range of different subjects. Although there is no current experimental evidence to indicate any manifestation of supersymmetry in nature, it is the case that models possessing such novel symmetry have interesting properties. Among other things, it relates particles with different spins and goes over the powerful constraints imposed by the Coleman–Mandula theorem [1]. To be precise, the only symmetries allowed in such a framework are those which turn out to be locally isomorphic to the direct product of an internal symmetry and the Poincaré group. It is the case that supersymmetry offers a way to have unification of space–time and internal symmetries by going over the so-called graded groups. Despite the unfamiliar properties of the graded Lie algebras, which only close under a combination of commutation and anti-commutation relations, it soon became clear that supersymmetry could play an important role in most branches of theoretical physics.

To the best of our knowledge Nicolai was the first to introduce super-symmetric quantum mechanics (susy qm for short) as an illustrative example for the occurrence of supersymmetry in non-relativistic models [2]. Years later Witten realized that susy qm could provide itself a simple model to describe a suitable mechanism in order for supersymmetry to break spontaneously [3]. Unless otherwise noted we restrict ourselves in the following to one-dimensional models. Whenever the

so-called superpotential function has no singularities along the whole domain $-\infty < x < \infty$, the susy qm models can be classified into the case where supersymmetry appears either exact or spontaneously broken. To be more specific, if susy remains unbroken the energy of the non-degenerate ground state is zero while the spontaneous breaking translates into a non-unique ground state with positive energy. In addition, several authors have conjectured the possibility that instantons could provide non-perturbative scenarios for the symmetry breaking phenomenon [3, 4].

On the other hand, it is a well-known fact by now that the transformation of a linear differential equation of second order to the Riccati equation form represents a common tool for both susy qm and the old factorization procedure in quantum physics. It may be interesting at this point to recall that the factorization method was introduced by Schrödinger to solve the hydrogen atom by means of purely algebraic arguments [5]. When Witten shows in susy qm as a toy model in quantum field theory it became soon clear that the realization of the superalgebra (graded Lie algebra in the mathematical jargon) in terms of first-order differential operators corresponds to a modern reformulation of the factorization procedures.

Let us now to outline the connection between susy qm and the harmonic oscillator which is perhaps the most studied potential for both the classical and quantum problems. It is the case that the old factorization method was first used to determine the spectrum of the one-dimensional harmonic oscillator [6]. In recent years, however, the possibility of obtaining general families of isospectral hamiltonians by means of a generalization of the factorization method has been realized. Going to the harmonic oscillator, Mielnik [7] was the first to incorporate in the framework of susy qm the set of anharmonic potentials previously analyzed by Abraham and Moses [8] using the Gelfand–Levitan formalism. Once the well-known algebra of raising (creation) and lowering (annihilation) is rewritten in the language of susy qm, the occurrence of the Riccati equation enables Mielnik to find a one-parameter family of potentials whose spectra are identical to that of the harmonic oscillator except for the lowest eigenvalue [7].

There is another solvable one-dimensional model which shares several properties with the harmonic oscillator. We refer to the so-called isotonic oscillator¹, one of the most used models for the study of the dynamics of nonlinear systems [9,10]. In particular the energy spectrum of this potential becomes isomorphous to the harmonic oscillator spectrum because it includes an infinite set of equidistant levels.

Returning to susy qm, apart from a few exceptions most of the papers in the subject deal with regular superpotentials so that the role of the singular ones has not been investigated in detail. To the best of our knowledge, Jevicki and Rodrigues [11] were the first to introduce in susy qm singular superpotentials. From a physical point of

¹ Also known as the *harmonic oscillator with centripetal barrier* since it is related to the radial equation for the three-dimensional harmonic oscillator.

view, singular superpotentials are responsible for negative energy eigenstates and partial degeneracy between the spectra of susy partner hamiltonians. More on this later.

In this article we set up a novel supersymmetric connection between the harmonic oscillator and the isotonic one. For such an objective, we take advantage of the non-standard factorization method which uses as reference the energy of the first excited level of the harmonic oscillator. On the other hand, we manage to obtain a one-parameter family of potentials whose spectra coincide with that of the isotonic oscillator.

To start from scratch, let us consider the form of the density lagrangian L_s which describes the occurrence of supersymmetry in non-relativistic quantum mechanics [2, 3]

$$L_s = \dot{x}^2 - W^2(x) + (\psi^* \dot{\psi} - \dot{\psi}^* \psi) - [\psi^*, \psi] W'(x), \quad (1)$$

where x represents as usual the even Grassmann coordinate associated with the bosonic degree of freedom; in addition, ψ and ψ^* denote fermionic degrees of freedom (odd Grassmann numbers) and fulfil $\{\psi, \psi^*\} = 0$, $\psi^2 = 0 = \psi^{*2}$. One could just as well consider (1) as the $(0+1)$ -dimensional version of a supersymmetric model in quantum field theory where the well-behaved function $W(x)$ represents the superpotential function. Resorting to the formalism of the superspace spanned by the coordinates t and two Grassmann variables η and η^* , the model defined by (1) is invariant under the supersymmetric transformations [1, 2]

$$\delta x(t) = \psi(t)\eta + \psi^*(t)\eta^*, \quad (2.a)$$

$$\delta \psi(t) = -[\dot{x}(t) + W(x)]\eta^*, \quad (2.b)$$

$$\delta \psi^*(t) = [\dot{x}(t) - W(x)]\eta, \quad (2.c)$$

As a matter of fact the invariance of (1) under the susy transformations described in (2) is obvious since

$$\delta L_s = (d/dt) [(\dot{x}(t) - W(x))\psi(t)\eta + (\dot{x}(t) + W(x))\psi^*(t)\eta^*]. \quad (3)$$

Going to the quantum level we can take advantage of the framework exposed in Refs. [3, 4], i.e.

$$\psi = \sigma^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad (4.a)$$

$$\psi^* = \sigma^- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad (4.b)$$

so that the susy qm hamiltonian H_s operates on two-component wave functions like

$$\varphi(x) = \begin{pmatrix} \varphi_+(x) \\ \varphi_-(x) \end{pmatrix} \quad (5)$$

according to its diagonal form, i.e.,

$$H_s = \begin{pmatrix} H_+ & 0 \\ 0 & H_- \end{pmatrix} \quad (6)$$

where as usual

$$H_{\pm} = -(\mathrm{d}^2/\mathrm{d}x^2) + W^2(x) \pm W'(x). \quad (7)$$

Notice that $H_{\pm}$ can be written in terms of the supercharges operators Q and Q^* with

$$Q = (\mathrm{d}/\mathrm{d}x) + W(x), \quad (8.a)$$

$$Q^* = -(\mathrm{d}/\mathrm{d}x) + W(x), \quad (8.b)$$

so that the conventional formalism of susy qm is recovered, i.e.

$$H_- = Q^*Q, \quad (9.a)$$

$$H_+ = QQ^*. \quad (9.b)$$

As long as the supercharges themselves fulfil the adjointness conditions $Q^\dagger = Q^*$ and $(Q^*)^\dagger = Q$, the spectrum of H_s is bounded from below by zero and states $\varphi_-(x)$ and $\varphi_+(x)$ with positive energy appear degenerate. In addition, the supercharges annihilate the ground state of the model in order for susy to remain unbroken. In other words

$$Q|\varphi_{0-}\rangle = Q^*|\varphi_{0+}\rangle = 0 \quad (10)$$

so that the zero-energy eigenstates $\varphi_{0\pm}(x)$ are obtained by solving first-order differential equations, i.e.

$$\varphi_{0\pm}(x) \sim \exp\left(\pm \int^x W(z) \mathrm{d}z\right). \quad (11)$$

As square integrable zero-energy modes demand boundary conditions like $\varphi_{0\pm} \rightarrow 0$ as $x \rightarrow \pm \infty$, the meaning of (11) is clear: the zero-mode, if it exists, must be non-degenerate.

Let us outline now the way in which the old factorization method takes shape in the new framework of susy qm. In such a case the goal is to represent the hamiltonian H associated with the one-dimensional potential $V(x)$, i.e.

$$H = -(\mathrm{d}^2/\mathrm{d}x^2) + V(x) \quad (12)$$

in terms of first-order differential operators. Taking advantage of the super charges Q and Q^* , we can start from the ansatz

$$H = Q^*Q + \varepsilon, \quad (13)$$

where ε represents the so-called factorization energy. In accordance with (12) and (13) we can establish the identification

$$V(x) = W^2(x) - W'(x) + \varepsilon \quad (14)$$

so that the superpotential $W(x)$ appears as the solution of a Riccati equation [12]. Without loss of generality the best way of solving this type of equations is as follows: starting from a particular solution of (14), say $W_p(x)$, the general one $W_g(x)$ can be obtained writing $W(x) = W_p(x) + W_g(x)$. This yields

$$W_g'(x) - 2W_p(x)W_g(x) - W_g^2(x) = 0. \quad (15)$$

One could just as well consider now a new function like $y(x) = 1/W_g(x)$, thus ending with a first-order linear differential equation, namely

$$y'(x) + 2W_p(x)y(x) + 1 = 0, \quad (16)$$

which gives rise to the formal solution

$$y(x) = \left[C - \int^x \exp\left(\int^t 2W_p(z) dz\right) dt \right] \exp\left(-\int^x 2W_p(z) dz\right) \quad (17)$$

where $C \in \mathcal{R}$. Among other things, we can write now the susy partner H_p associated with H , i.e.

$$H_p = QQ^* + \varepsilon, \quad (18)$$

a hamiltonian characterized by the potential $V_p(x)$, namely

$$V_p(x) = V(x) + 2\frac{d}{dx}\left(W_p(x) + \frac{\exp(\int^x 2W_p(z) dz)}{C - \int^x \exp(\int^t 2W_p(z) dz) dt}\right). \quad (19)$$

It is worth recalling at this point that different choices can be made for the so-called factorization energy ε [13]. Almost from the very beginning of the subject, most of the papers deal with the regular superpotential $W_p(x)$ derived from the ground state $\psi_0(x)$ of H . In other words, the factorization energy ε equals the ground state energy E_0 . In order not to clutter the paper, we simply sketch the main results obtained from such a choice. First of all, the logarithmic derivative of the ground state provides us with the particular solution $W_p(x)$ of the Riccati equation (see (14)). In short

$$W_p(x) = -\psi_0'(x)/\psi_0(x). \quad (20)$$

In addition, the superpotential $W_p(x)$ is regular along the whole domain $-\infty < x < \infty$ because the ground state $\psi_0(x)$ has no zeros. As regards the specific realization of supersymmetry the scheme is as follows:

(i) The ground state $\psi_0(x)$ of H plays the role of the zero-energy mode of $H_- = Q^*Q$ (notice that $H = H_- + E_0$) whenever the operators Q and Q^* include the superpotential $W_p(x)$.

(ii) Once $W_p(x)$ is obtained, it suffices to solve (15) in order to get the one-parameter

family of superpotentials $W(x)$. The whole procedure is completed when writing partner hamiltonian H_p .

(iii) Supersymmetry itself remains exact so that the hamiltonians H and H_p share the spectra except for the ground state $\psi_0(x)$ of H . In summary, the above discussion allows us to find a one-parameter family of isospectral hamiltonians H_p . To make clear the relationship between the spectra of H and H_p it seems plausible to take the limit $|C| \rightarrow \infty$ in H_p . More on this later.

The question we wish to address in the following is to show how the factorization method works when considering the harmonic oscillator. Proceeding along the lines established above we start from the hamiltonian

$$H = -(\mathrm{d}^2/\mathrm{d}x^2) + x^2 \quad (21)$$

which gives rise to the well-known eigenfunctions [6]

$$\psi_n(x) \sim H_n(x) \exp(-x^2/2), \quad (22)$$

where $H'_n(x)$ represent Hermite polynomials which in turn can be written in terms of the Rodriguez formula

$$H_n(x) = (-1)^n \exp x^2 (\mathrm{d}^n/\mathrm{d}x^n) \exp(-x^2), \quad (23)$$

As regards the energy levels E_n associated with the eigenfunctions $\psi_n(x)$ we have

$$E_n = 2n + 1, \quad n = 0, 1, 2, \dots \quad (24)$$

Taking advantage now of the form of the ground state function $\psi_0(x)$ in (22) we have (see (20))

$$W_p(x) = x \quad (25)$$

so that the superpotential $W(x)$ reads

$$W(x) = x + \frac{\exp x^2}{C - \int^x \exp t^2 \mathrm{d}t}. \quad (26)$$

In such a case we get finally a one-parameter family of isospectral potentials $V_p(x)$ like

$$V_p(x) = x^2 + 2(\mathrm{d}/\mathrm{d}x) \left(\frac{\exp x^2}{C - \int^x \exp t^2 \mathrm{d}t} \right), \quad (27)$$

in the spirit of the previous results obtained by Mielnik [7] using non-standard factorizations of the harmonic oscillator. For one thing, the constant C must be chosen in order for the superpotential $W(x)$ to avoid a singularity along the real axis. On the other hand, the relationship between the spectra of H and H_p is clear when considering the limit $|C| \rightarrow \infty$ in (27). Notice the isospectral character of all the potentials included in $V_p(x)$. To sum up, we find in such a limit

$$H_p = -(\mathrm{d}^2/\mathrm{d}x^2) + x^2 + 2, \quad (28)$$

which gives rise to the energy levels

$$E_n = 2n + 3, \quad n = 0, 1, 2, \dots \quad (29)$$

so that one can check the isospectral character of H and H_p except for the ground state of the proper H .

Returning to susy qm, except for a few exceptions most of the papers deal with regular superpotentials $W(x)$ along the wide domain $-\infty < x < \infty$. However, in fact a general method for the generation of singular superpotentials starting from the first excited level of any symmetric one-dimensional $V(x)$ has been recently introduced [14,15]. Among other things, the singular superpotentials are responsible for the occurrence of negative energies and the partial degeneracy between the spectra of the partner hamiltonians. We shall return on these subjects later on. As a matter of fact the above discussion is borne out by the pioneer work of Jevicki and Rodrigues [11]. These authors consider a superpotential $W(x)$ of the form

$$W(x) = g/x - x \quad (30)$$

and analyse the different realizations of susy depending on the values of the coupling constant q . For the sake of brevity, we simply outline the relationship between the occurrence of singular superpotentials and the factorization of the hamiltonian H which appears in (12), whenever the choice for the so-called factorization energy is E_1 , i.e. the energy of the first excited level of $V(x)$. Writing therefore

$$H = Q^*Q + E_1 \quad (31)$$

it is the case that (14) provides us with the Riccati equation

$$V(x) = W^2(x) - W'(x) + E_1. \quad (32)$$

In order to find the particular solution $W_p(x)$, it suffices again to have the logarithmic derivative of the eigenstate associated with the factorization energy. To sum up

$$W_p(x) = -\psi'_1(x)/\psi_1(x). \quad (33)$$

As we are dealing with the first excited eigenstate of $V(x)$, the eigenfunction $\psi_1(x)$ has only one zero, say at $x = x_s$. In the neighborhood of such a point, the proper superpotential becomes singular since

$$\psi_1(x) = c(x - x_s), \quad (34.a)$$

$$W_p(x) = -1/(x - x_s), \quad (34.b)$$

so that the partner potential $V_p(x)$ reads

$$V_p(x) = 2/(x - x_s)^2. \quad (35)$$

Let us explain at this point the way in which a singular superpotential like (34.b) can break the whole domain $-\infty < x < \infty$ into two disjoint regions as regards the partner potential $V_p(x)$. (Remember the regular character of the starting potential $V(x)$.) Without loss of generality, we analyse the behaviour of

$$V(x) = \lambda/(x - x_s)^2 \quad (36)$$

in terms of the values of the parameter λ [16,17]. As a matter of fact, three different regions appear, namely:

(i) If $\lambda \in (-\infty, -1/4)$ the model becomes unphysical since the spectrum is not bounded from below [16].

(ii) When $\lambda \in (-1/4, 3/4)$ the singularity is not strong enough to make the eigenfunctions vanish at $x = x_s$. However, it is necessary to define a *one-parameter family of self-adjoint extensions* so that a spectrum bounded from below appears once a proper domain is specified [16]. In addition, the wave functions pass across the singularity point $x = x_s$ and the model extends itself along $-\infty < x < \infty$.

(iii) Finally, if we consider the range $\lambda \in (3/4, \infty)$ the singularity acts as an impenetrable barrier, thus dividing the space into two independent regions, i.e. $x \leq x_s$ and $x > x_s$. Going to more physical terms, the wave functions must vanish at $x = x_s$ so that the lack of communication between the two pieces of space is absolute [16,17].

Returning to our susy qm framework, it is clear from (34.b) and (35) that we face a model where the partner potential $V_p(x)$ exhibits strong singularity when $x = x_s$ so that the eigenfunctions must vanish at such a point. On the other hand, the singular character of the superpotential brings difficulties when making the connection between the eigenstates of H and H_p . If these features are not taken into account, one encounters apparent paradoxes as regards the susy realization. For one thing, the most charming properties of susy qm are due to the supersymmetry algebra acting as an algebra of formal differential operators. However, such properties (the positivity of the energy and the pairing of states) are valid only if the above mentioned algebra holds effectively as an operator algebra in Hilbert space. It seems plausible therefore to establish a connection between the specific realization of susy and the conditions imposed on admissible states [18,19]. Going to more physical terms, the Hilbert space associated to $V_p(x)$ must be narrowed as compared to the Hilbert space of the hamiltonian with potential $V(x)$. In such a case, susy realizes itself on eigenstates which vanish at $x = x_s$ so that only partial degeneracy must be expected at best. The occurrence of negative energies in susy qm can be explained as follows: when factorizing in terms of the first excited level $\psi_1(x)$, the ground state of the operator Q^*Q acquires negative energy in order for the global hamiltonian H (notice that $H = Q^*Q + E_1$) to start its spectrum from E_0 .

Let us write down the way in which the above discussion takes shape when considering the harmonic oscillator. Starting now from the first excited level $\psi_1(x)$ (see (22) and (23))

$$\psi_1(x) \sim x \exp(-x^2/2), \quad (37)$$

we get for $W_p(x)$

$$W_p(x) = x - 1/x. \quad (38)$$

Resorting as usual to the Riccati equation we have

$$W(x) = x - \frac{1}{x} + \frac{\exp x^2}{x(Cx - x \int^x t^{-2} \exp t^2 dt)} \quad (39)$$

so that the partner potential $V_p(x)$ reads

$$V_p(x) = x^2 + \frac{2}{x^2} + 2 + \frac{d}{dx} \left(\frac{\exp x^2}{x(Cx - x \int^x t^{-2} \exp t^2 dt)} \right). \quad (40)$$

At this point it is worth noting different features of the result we have obtained, namely:

(i) First of all, the constant C must be chosen now in order for the superpotential $W(x)$ to avoid a second singularity which could go with the one already existing at $x = 0$.

(ii) Due to the isospectral character of all the potentials $V_p(x)$, we can take advantage of the limit $|C| \rightarrow 0$ before making the comparison between the spectra of H and H_p . In such a case we have

$$H_p = -\frac{d^2}{dx^2} + x^2 + \frac{2}{x^2} + 2, \quad (41)$$

so that as a matter of fact the potential which appears in (40) represents a one-parameter family of models with the spectrum of the isotonic oscillator².

(iii) As $V_p(x)$ exhibits a strong singularity at the origin, the space breaks up into two disjoint regions ($x \geq 0$ or $x \leq 0$) without communication between them since the wave functions vanish at $x = 0$.

(iv) Restricting ourselves to the half-line ($x \geq 0$), the wave functions $\psi_n(x)$ of (41) can be written as [17,20]

$$\psi_n(x) \sim x^2 \exp(-x^2/2) M_n(-n, 5/2, x^2), \quad n = 0, 1, 2, \dots \quad (42)$$

where the M'_n represent Kummer confluent hypergeometric functions.

(v) As regards the spectrum of (41) we have

$$E_n = 4n + 7, \quad n = 0, 1, 2, \dots \quad (43)$$

When comparing (43) with (24), we conclude that H_p skips the first three levels of H . To be precise, susy realizes itself acting on the eigenfunctions of H which vanish at $x = 0$. In such a case only the odd wave functions (excepting the zero-energy mode of Q^*Q) of the Hilbert space of H can be transported via supersymmetry into the Hilbert

² We ignore trivial shifts of the origin since we are concerned only with different shapes of $V(x)$ as a function of x .

space associated with H_p . In addition, the occurrence of negative energies is not surprising whenever the factorization is carried out in terms of the first excited level eigenvalue E_1 . Such a choice pushes the ground state of Q^*Q to negative energies in order for H to start its spectrum from E_0 .

Proceeding along the same lines, one could just as well consider the factorization method when using the second excited level $\psi_2(x)$ of the harmonic oscillator as reference, i.e.

$$\psi_2(x) \sim (2x^2 - 1) \exp(-x^2/2) \quad (44)$$

which gives rise to

$$W_p(x) = x - 4x/(2x^2 - 1), \quad (45)$$

a superpotential with strong singularities located at $x = \pm 1/\sqrt{2}$, so that the real axis breaks itself into three disjoint regions and the problem concerning $V_p(x)$ becomes intractable.

To finish we would like to stress the following points:

(i) Taking advantage of the non-standard factorization method which chooses as reference the first excited level of the harmonic oscillator, we manage to establish a novel supersymmetric connection between the proper harmonic oscillator and the isotonic one. To be precise, we obtain a one-parameter family of potentials with the spectrum of the isotonic oscillator.

(ii) Supersymmetry appears neither exact nor spontaneously broken but the Hilbert space associated to the models with $V_p(x)$ is narrowed as compared to the Hilbert space of the hamiltonian with $V(x)$.

(iii) The above statements are borne out by the following fact: susy realizes itself acting on the wave functions of $V(x)$ which vanish at $x = 0$. Going to more physical terms, only the *half-spectrum* is transferred to the partner potential.

(iv) In addition, while the domain of the hamiltonian H (harmonic oscillator) extends along the whole real axis $-\infty < x < \infty$, the partner H_p reduces itself to the half-line ($x \geq 0$ or $x \leq 0$).

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